

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	[A particle's] maximum displacement [from the equilibrium position] Accept distance from {crest / trough} to {midpoint / rest / equilibrium point} Don't accept maximum height of wave if unqualified	1			1		
		(ii)	$f = 50 \text{ [Hz]}$ or $T = 0.02 \text{ [s]}$ (1) Use of $v = f\lambda$ or $v = \frac{\lambda}{T}$ (1) $= 7.5 \text{ m s}^{-1}$ or 750 cm s^{-1} unit mark (1)	1	1 1		3	3	
		(iii)	Sinusoidal wave with same A and T as original (1) Starting at (0,6) i.e. cos function drawn (1) Minimum of 1 wave drawn		2		2	2	
	(b)		Indicative content: Formation of stationary wave: - two [progressive] waves travelling in opposite directions - same frequency / wavelength / amplitude - reference to pulley / fixed point to act as reflector - interference / superposition - nodes and antinodes Stationary / progressive comparison: - Stationary waves have the same frequency or wavelength as the progressive waves which created them - Particles oscillate in both - ENERGY: not transferred (S) / transferred (P) - PHASE: neighbouring particles [in same loop] in phase (S) / neighbouring particles not in phase / continuous change of phase along the wave (P)	6			6		